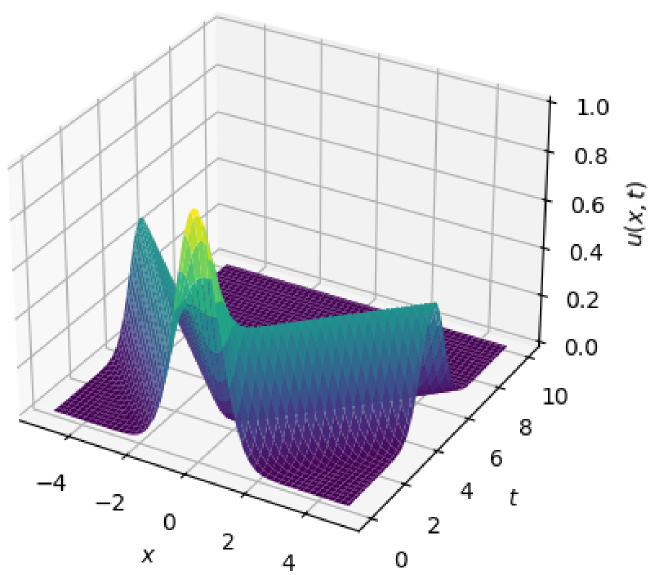
Question 1

(a) Given $\phi(x) = e^{-x^2},\ \psi(x) = 0$

$$\begin{aligned} u(x,t) &= \frac{1}{2}\left(e^{-(x-ct)^2} + e^{-(x+ct)^2}\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} 0 ds \\ &= \frac{1}{2}\left(e^{-(x-ct)^2} + e^{-(x+ct)^2}\right) \end{aligned}$$

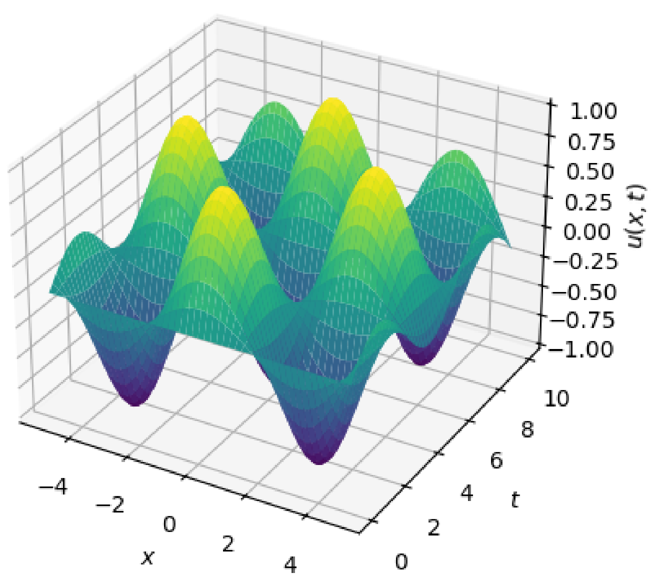
D'Alembert soln for $\phi(x) = e^{-x^2}$ and $\psi(x) = 0$ with $c = 1$



(b) Given $\phi(x) = 0,\ \psi(x) = \cos x$

$$\begin{aligned} u(x,t) &= \frac{1}{2}\left(0+0\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \cos(s)ds \\ &= \frac{1}{2c}\Big[\sin(s)\Big]_{x-ct}^{x+ct} \\ &= \frac{1}{2c}\Big[\sin(x+ct) - \sin(x-ct)\Big] \end{aligned}$$

D'Alembert soln for $\phi(x) = 0$ and $\psi(x) = \cos x$ with $c = 1$



(c) Given $\phi(x) = \begin{cases} 1 & |x| < 1 \\ 0 & \text{otherwise} \end{cases}, \quad \psi(x) = 0$

For $|x| < 1$,

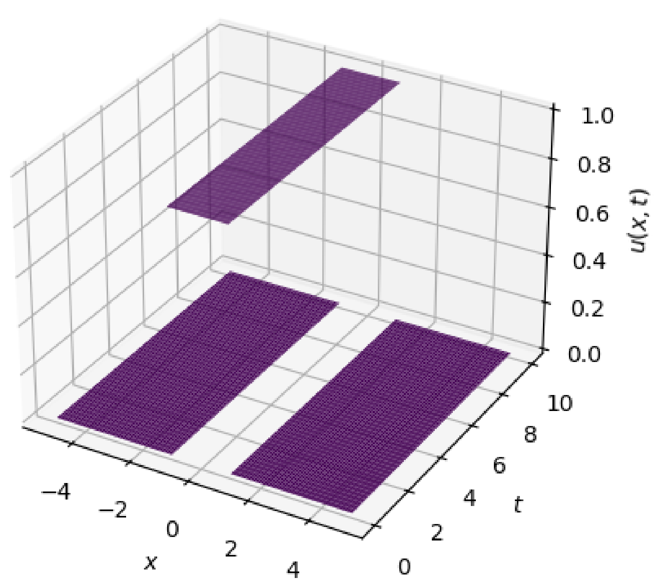
$$\begin{aligned} u(x,t) &= \frac{1}{2}(1+1) \\ &= 1 \end{aligned}$$

For $|x| > 1$

$$u(x,t) = 0$$

$$u(x,t) = \begin{cases} 1 & |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

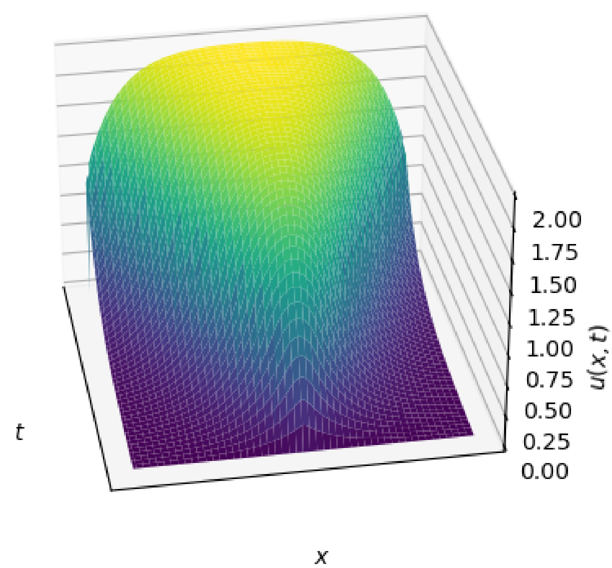
D'Alembert soln for $\phi(x) = 1$ if $|x| < 1$ and 0 otherwise, and $\psi(x) = 0$ with $c = 1$



(d) Given $\phi(x) = 0,\ \psi(x) = e^{-|x|}$

$$
\begin{aligned}
u(x,t) &= \frac{1}{2c}\int_{x-ct}^{x+ct} e^{-|s|}ds \\
&= \begin{cases} \frac{1}{2c}\int_{x-ct}^{x+ct} e^{s}ds & s<0 \\ \frac{1}{2c}\int_{x-ct}^{x+ct} e^{-s}ds & s>0 \end{cases} \\
&= \begin{cases} \frac{1}{2c}\Big[e^{s}\Big]_{x-ct}^{x+ct} & s<0 \\ \frac{1}{2c}\Big[-e^{-s}\Big]_{x-ct}^{x+ct} & s>0 \end{cases} \\
&= \begin{cases} \frac{1}{2c}\Big[e^{x+ct} - e^{x-ct}\Big] & x<-ct \\ \frac{1}{2c}\Big[-e^{-(x+ct)} - e^{(x-ct)}\Big] + \frac{1}{c} & -ct<x<ct \\ \frac{1}{2c}\Big[-e^{-(x+ct)} + e^{-(x-ct)}\Big] & x>ct \end{cases}
\end{aligned}
$$

D'Alembert soln for $\phi(x) = 0$ and $\psi(x) = e^{-|x|}$ with $c = 1$



Question 2

(a) Given

Question 3
Question 4
Question 5